

# Radwa Amr Helmy Hafez

Computer Engineering Student | Data Science & Machine Learning Enthusiast

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## 🎓 Education

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| <b>Bachelor's Degree in Computer Engineering,</b><br><i>Helwan University – Faculty of Engineering, Computer and Systems Department</i><br>(Expected Graduation: June 2027) | 09/2022 – Present<br>Cairo, Egypt |
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## 🧩 Professional Summary

Computer Engineering student at Helwan University specializing in Data Science and Machine Learning, with hands-on experience in data analysis, visualization, and building ML models using Python, Scikit-learn, and Pandas. Trained across multiple programs including DEPI, ITI, and NTI, building a strong foundation in supervised and unsupervised learning and data preprocessing.

## 👜 Professional Experience & Internships

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| <b>Data Scientist Training, Digital Egypt Pioneers Program (DEPI),</b><br><i>Ministry of Communications and Information Technology (MCIT)</i> | 06/2025 – 12/2025<br>Egypt |
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- Completed a 6-month intensive Data Scientist track covering the full data science pipeline from methodology to deployment
- Built hands-on projects using **Python** for data analysis, visualization, and machine learning
- Worked with **databases and SQL** for data querying and management alongside Python integration
- Applied **data analysis and visualization** techniques using libraries such as Pandas, Matplotlib, and Seaborn
- Implemented **machine learning models** using Scikit-learn covering supervised and unsupervised learning

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| <b>AI Summer Training, Information Technology Institute (ITI)</b> | 08/2025 – 09/2025 |
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- Covered AI fundamentals, probability & statistics, linear algebra, and numerical optimization for ML.
- Practiced Python for data preparation & exploration.
- Gained knowledge of neural networks and deep learning basics

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| <b>Machine Learning Summer Training, National Telecommunication Institute (NTI)</b> | 07/2025 – 08/2025 |
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- Completed 120 hours (90h technical + 30h Freelancing & Career Building), scored 99%.
- Hands-on training in data preprocessing, feature engineering, model validation, and evaluation.
- Implemented ML algorithms: Regression, Naive Bayes, KNN, SVM, Random Forest, Gradient Boosting, XGBoost.
- Explored unsupervised learning (K-means, GMM, PCA)

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| <b>AI/ML Summer Intern, Faculty of Engineering, Helwan University</b> | 07/2024 – 08/2024 |
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- Applied Python, NumPy, Pandas, and Matplotlib for data preprocessing & visualization.
- Practiced OOP, EDA, and supervised/unsupervised learning.
- Built a predictive maintenance model using Kaggle sensor data with Random Forest & Neural Networks

## Projects

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### **Sales Forecasting & Optimization System**, *Python, ARIMA, DVC, Pandas, Streamlit, MLflow*

- Built a time-series forecasting system on a real supermarket dataset (3M+ rows) merging sales, store, holiday, and oil price data
- **Personally responsible for** data preparation, cleaning, preprocessing, and ARIMA model development
- Collaborated with a team of 6 Data Scientists to compare forecasting models (Prophet, XGBoost, LightGBM, SARIMAX) and deploy the final system as an interactive Streamlit dashboard
- Implemented MLOps practices including model monitoring, data versioning via DVC, and automated email alerts for prediction drift

### **Network Intrusion Detection System**, *Python, Scikit-learn, XGBoost, Pandas*

- Built a binary classifier on the UNSW-NB15 dataset to detect network attacks vs. normal traffic
- Applied full ML pipeline: EDA, preprocessing, encoding, scaling, and model training across 6 algorithms (LR, DT, RF, KNN, NB, XGBoost)
- Improved final performance using an ensemble Voting Classifier

### **Breast Cancer Detection**, *Python, Scikit-learn, Pandas, Seaborn*

- Classified tumors as malignant or benign using the Wisconsin Breast Cancer dataset
- Trained and compared multiple SVM kernels (linear, RBF, polynomial, sigmoid) with hyperparameter tuning via GridSearchCV
- Achieved ~98% accuracy with cross-validated ROC-AUC evaluation

### **Mushroom Classification**, *Python, Scikit-learn, Pandas, Seaborn*

- Predicted mushroom edibility (edible vs. poisonous) from physical and ecological features
- Handled data cleaning, feature encoding, and class balance analysis before modeling
- Evaluated 4 classifiers using Accuracy, Precision, Recall, F1, ROC-AUC, and Cross-Validation

### **Heart Disease Prediction**, *Python, Scikit-learn, Pandas, Matplotlib, Seaborn*

- Analyzed clinical features to predict heart disease presence and assess patient risk
- Performed full data preparation, cleaning, preprocessing, and exploratory visualization
- Applied classification and regression modeling with comprehensive evaluation pipeline

## Technical Skills

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### **Programming**

- Python, C++, Java, OOP, Data Structures

### **Visualization**

- Matplotlib, Seaborn, Plotly

### **Tools & Environments**

- Jupyter Notebook, Google Colab, VS Code

### **Data Analysis**

- NumPy, Pandas, Data Preprocessing, Feature Engineering, EDA

### **Deep Learning**

- Neural Networks (Keras, TensorFlow – basic)

### **Machine Learning**

- **Supervised:** Classification (Naive Bayes, KNN, SVM, Decision Trees, Random Forest, Gradient Boosting, XGBoost), Regression
- **Unsupervised:** K-means, PCA, Gaussian Mixture Models (GMM)

### **Databases**

- SQL, MySQL

## Languages

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### **Arabic**

Native

### **English**

Advanced

### **German**

Basic